

CH1101: Quantum Chemistry and Spectroscopy

Tutorial Sheet 04

Module: Quantum Chemistry

- Q1. Consider a free particle constrained to move over the rectangular region $0 \leq x \leq a$, and $0 \leq y \leq b$. The eigenfunctions of this system are

$$f_{n_x} f_{n_y}(x, y) = \left(\frac{4}{ab}\right)^{1/2} \sin \frac{n_x \pi x}{a} \sin \frac{n_y \pi y}{b}$$

where, $n_x = 1, 2, 3, \dots$ and $n_y = 1, 2, 3, \dots$; the Hamiltonian operator for this system is

$$\hat{H} = -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right)$$

Find the energy eigenvalues.

- Q2. Specify which of the following operators are Hermitian: $-i \frac{d}{dx}$, $\frac{d^2}{dx^2}$, and $i \frac{d^2}{dx^2}$. Assume, that $0 \leq x \leq L$, and the functions on which these operators operate are appropriately well behaved:

$$(i) f(x) = \sin \frac{\pi x}{L}, \quad \text{where } f(0) = f(L) = 0$$

$$(ii) f(x) = e^{ix/L}, \quad \text{where } f(0) = f(L)$$

- Q3. The wavefunction of an electron in the lowest energy state of a He^+ ion is proportional to e^{-2r/a_0} , where r is the distance from the nucleus. Calculate the relative probabilities of finding the electron inside a small cubic volume located at a distance: (a) $r = \frac{a_0}{2}$, and (b) $r = a_0$.

